Total number of printed pages–3

5 SEM MCA (NCBCS) 4

2015

(June)

ARTIFICIAL INTELLIGENCE

Paper : 504

Full Marks : 75

Pass Marks : 34

Time : Three hours

The figures in the margin indicate full marks for the questions.

Answer Q. No. ***1*** *and* ***any four*** *from the rest.*

1. *(a)* Show that $\sim p(\vee(\sim p \wedge q)$ and $\sim p \wedge \sim q$ are logically equivalent. 4

   (b) Prove whether tautologies or contradiction for the following 6

   (i) $(P \rightarrow Q) \leftrightarrow (\sim P \vee Q)$

   (ii) $(p \vee q) \wedge \sim(\sim p \wedge (\sim q \vee \sim r)))$
   $\vee (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$

Contd.

(c) What is semantic net? 1

(d) What is the aim and scope of Artificial Intelligence? 4

2. (a) Write down the basic ways to perform a search. Briefly describe Hill Climbing. 6

(b) What do you mean parallel and distributed A-I? 4

(c) Briefly describe about Robotics. 5

3. (a) What is the difference between generate and Test method and means-end-analysis method? 5

(b) Briefly describe Rule-Base system. 6

(c) What is IS-A and instance relationship? 4

4. (a) What are the characteristics of Blind and heuristic search? 4

(b) What is conceptual dependency? How it can be used to represent as knowledge? 4

(c) Briefly explain firing rule for neural network, Biological neural network and artificial neural network. 7

5. (a) What is Expert System? Brie describe the components of Exp System. What are the criteria adopting an Expert System? 4

(b) What is learning and chaining?

(c) What is monotonic, nonmonotonic default and Fuzzy Logic?

6. (a) What do you mean Resolution refutation?

(b) Describe the function of α–β pruning

(c) Write down the basic components o propositional logic.

(d) What is knowledge, Events & Domain experts?

7. Write short notes on : **(any three)** 3×5=15

(1) Image processing & night vision technique

(2) Frames & Scripts

(3) Dandral

(4) AND-OR-Graph

(5) Differentiate Human Expert and Artificial Expert.

300

5 MCA (NC) AI 4

2013

(December)

ARTIFICIAL INTELLIGENCE

Paper : 504

Full Marks : 75

Pass Marks : 34

Time : Three hours

The figures in the margin indicate full marks for the questions.

*Answer **Q. No. 1** and **any four** from the rest.*

(a) State true *or* false. 5

(i) Form $P \rightarrow Q$, infer $\sim Q \rightarrow \sim P$.

(ii) Default logic is a non-monotonic reasoning.

(iii) MYCIN is a expert system.

Contd.

(iv) alpha-beta pruning is not a sear algorithm.

(v) Reasoning infer facts from existi data.

(b) What do you mean A.I. ?

(c) Show that

$$(P \Rightarrow Q) \wedge (R \Rightarrow Q) = (P \vee R) \Rightarrow Q \ ?$$

(d) If it is not rain and I have time then I w go to movie. Translate into propagatio logic.

(e) Write down the basic ways to perform search ?

2. (a) What is Expert System ? Describe t components of Expert System. What are t advantages and disadvantages of an Exp System ? 8

(b) Prove whether tautologies or contraditio for the following.

(i) $(P \wedge (P \rightarrow Q)) \rightarrow Q$

(ii) $(P \rightarrow Q) \leftrightarrow (\sim P \vee Q)$

(c) What are the limitation of hill climbin functions ? 2

(a) Briefly describe the function of Generate an test method. 5

(b) What do you mean Resolution-refutation ? 4

(c) List the advantages and disadvantages o Decision table ? 3

(d) What are the basic components o propositional logic ?

4. (a) Briefly describe forward and backwar chaining ?

(b) What is heuristic search, fuzzy logic, an default logic ?

(c) How can knowledge be represented usi conceptual dependency ?

5. (a) Briefly describe α-β pruning algorithm.

(b) What is Frames ? What are the advantage and disadvantages of scripts ?

(c) What is semantic net ?

6. (a) What is neural network ? Explain Biologica neural network and Artificial neural network Write down the application areas of neura network. 3+5+2=1

(b) Briefly describe A* algorithm.

7. Write short notes on : *(any three)* 5×3=1

(i) Rule-Base system

(ii) AND-OR Graph

(iii) MYCIN

(iv) Means-end analysis

(v) Differentiate human expert and A.I. expert

Total No. of printed pages = 3

5 SEM MCA AI 4 (NCBCS)

2016

(June)

COMPUTER APPLICATION

Paper : MCA-504

(Artificial Intelligence)

(Elective-I)

Full Marks – 75

Pass Marks – 34

Time – Three hours

The figures in the margin indicate full marks for the questions.

Answer question No.1 and any *four* from the rest.

1. (a) What do you mean by Turing Test ?

(b) Briefly describe the goal of Artificial Intelligence.

(c) ISA and Instance relationship.

(d) Prove that

$X = ((P \Rightarrow Q) \wedge (Q \Rightarrow R)) \Rightarrow (P \Rightarrow R)$ is a tautology.

[Turn over

(e) What do you mean by parallel machine ? 3×5=15

2. (a) Briefly explain the Generate and Test method. 5

(b) What do you mean by reasoning ? Describe uncertain non-monotonic and default reasoning. 6

(c) Prove that 4

$$(P \Rightarrow Q) \wedge (R \Rightarrow Q) = (P \vee R) \Rightarrow Q$$

3. (a) What do you mean by A.I system ? What are the importance of A.I ? 6

(b) Briefly describe the properties of knowledge representation system. 5

(c) What is partial order planning ? 4

4. (a) Briefly describe mini-max algorithm. 6

(b) What is Expert system ? What are the criteria for adopting Expert system ? What are the advantages and disadvantages of Expert system ? Briefly describe the components of Expert system. 9

5. (a) How neural networks help us ? What are the rules of neural networks in Artificial Intelligence ? Briefly describe the firing rules of neural networks. 7

(b) Differentiate human expert and artificial expert. 6

(c) What is state space ? 2

6. (a) Briefly describe an algorithm. 6

(b) What are forward chaining , backward chaining and learning ? 6

(c) What do you mean by parallel and distributed A.I ? 3

7. Write short notes on any *three* : 5×3=15

(i) MYCIN

(ii) BFS and DFS

(iii) Robotics

(iv) Frames and scripts

(v) Fuzzy logic.

Total number of printed pages–3

5 SEM MCA (NCBCS) AI 4

2018

(June)

COMPUTER APPLICATION

Paper : 504

(Artificial Intelligence)

Full Marks : 75

Pass Marks : 34

Time : Three hours

The figures in the margin indicate full marks for the questions.

*Answer **any five** questions.*

1. a) What is A.I. ? Describe the application areas of A.I. 5

b) Prove De Morgan's law with the help of truth table. 5

c) Prove that
$(P \Rightarrow Q) \wedge (R \Rightarrow Q) = (P \vee R) \Rightarrow Q$ 5

Contd.

2. a) What do you mean by AND-OR graph? Describe AO* algorithm. 7

b) Write down the properties of Knowledge Representation System. 5

c) What is Planning? 3

3. a) Briefly describe Neural network. What are the advantages of neural network? 7

b) What do you mean by parallel and distributed A.I.? 5

c) How Artificial Intelligence is related to human-based nature? 3

. a) What do you mean by informed and uninformed search? 2

b) What is Expert system? What are the characteristics of Expert System? What are the advantages and disadvantages of Expert System? 7

c) What do you mean by Robotics? 6

5. a) Briefly describe the Rule-based system Architecture. 7

b) What do you mean by Natural-language processing? 5

c) What is Turing test? 2

d) What is Predicate logic? 1

6. a) What do you mean by heuristic search? Briefly explain Mini-Max algorithm. 7

b) What is Tautology and Contradiction? 2

c) Describe the Generate and Test method. 6

7. Write short notes on : ***(any three)*** 5×3=15

a) Hill Climbing

b) Frames and Scripts

c) MYCIN

d) Means-end analysis

e) Fuzzy logic.